

Candidate's Name: ..... Index No. ....

Signature: .....

545/1  
CHEMISTRY  
Paper 1  
April, 2026  
2hours

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|----------------------|--|--|
| Total weighted Score |  |  |
| Initials             |  |  |

## KABALE DIOCESE EXAMINATIONS

### Uganda Certificate of Education

CHEMISTRY  
PAPER 1

#### For Scorer's Use Only

#### INSTRUCTIONS TO STUDENTS:

- The paper consists of *two* Sections; *A* and *B*. It has six examination items.
- Section *A* has *two compulsory* items Responses to section *A* *must* be written in the spaces provided.
- Section *B* has two *Parts; I and II*. Respond to *only one* item from each part.
- Responses to section *B* *must* be written in the answer booklet(s) provided.
- Respond to *four* items in all.
- Responses to each item in Section *B* should start on a fresh page.
- Any additional item(s) responded to will *not* be scored.

| Item No.             | Basis code | Weighted Score | Scorer's Initials |
|----------------------|------------|----------------|-------------------|
| 1                    | T          |                |                   |
|                      | D          |                |                   |
|                      | E          |                |                   |
| 2                    | C          |                |                   |
|                      | X          |                |                   |
|                      | I          |                |                   |
| (3/4)                | P          |                |                   |
|                      | S          |                |                   |
|                      | B          |                |                   |
| _(5/6)               | N          |                |                   |
|                      | M          |                |                   |
|                      | Be         |                |                   |
| Total Weighted Score |            |                |                   |

## SECTION A

*Answer all items in this section in the spaces provided.*

### ITEM 1

When Hamza felt sick, his mother prepared an aloevera extract for him which provided a brief improvement on first day. However, his symptoms persisted the next day. He then went to a nearby facility where he was diagnosed with Gastrointestinal poisoning known as salmonella infection resulting from taking contaminated food. He was given a product which he used and recovered.

**Task.** As a chemistry student,

- (a) State the type of the product used which enabled him to completely recover.

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- (b) Advise him about any danger associated with the product he used.

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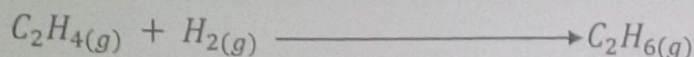
- (c) Evaluate the substances he used for treatment from the time he felt sick to the time of recovery.

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## Task 2

A production industry wants to convert ethene, an excess waste gas from its production processes into another useful product, ethane. They plan to react the hydrocarbon with hydrogen under suitable controlled conditions. Below is the equation of reaction.



The manager needs to know the mass of product in Kg that can be produced when 10,000 litres of ethene at *s. t. p* reacts with hydrogen. He also needs to know the environmental concern associated with the product. You have been invited for guidance

**Task;** As a learner of chemistry,

(a) Explain the category of each hydrocarbon in the reaction.

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(b) Suggest the suitability of product to its applications in everyday life.

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(c) Calculate the mass of product formed in above reaction ( $C = 12, H = 1$ )

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(d) Explain the environmental impact associated with the product.

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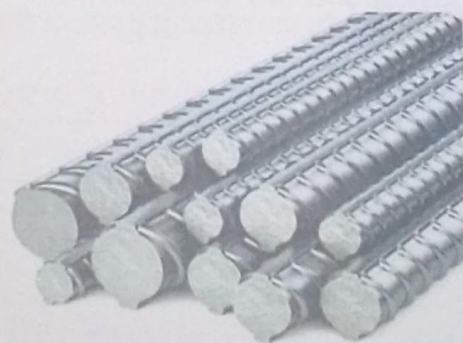
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## SECTION B

### Part I *(Attempt one item from this section)*

#### Item 3

The vast majority of iron is used to produce steel an alloy of iron and carbon used for making bars of construction, tools and heavy machinery. To increase on quantities of iron on market needed, the government has cleared an investor to set up an iron extraction plant in Rubanda. However, the locals are concerned about its impact and are seeking assurance on how they will benefit from its operations. The district officials have organized a workshop to educate the locals. You have been invited to attend.

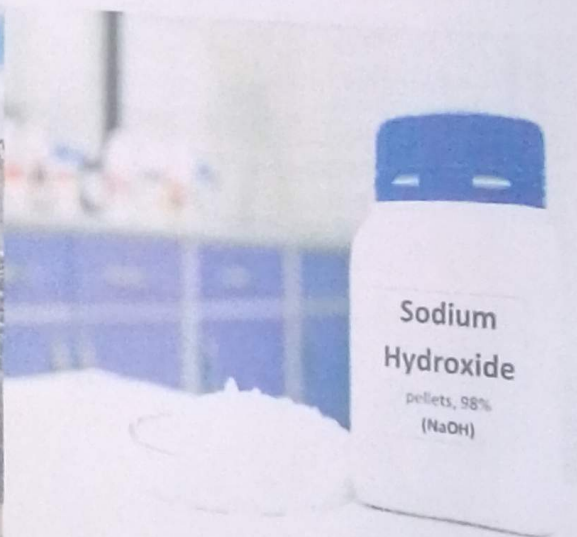


### Task.

As a chemistry student, make a write up you will use during the presentation.

### Item 4.

Due to large quantities of sodium hydroxide needed on market for production of detergents and bleaching agents, Jinja Municipal council officials have cleared an investor to set up a sodium hydroxide manufacturing plant. However, the locals are concerned about its impact to the environment and how they will socially benefit from the plant. The officials have organized workshop at the municipal hall. You have been invited to address the locals.



### Task.

Prepare a write up message you will use.



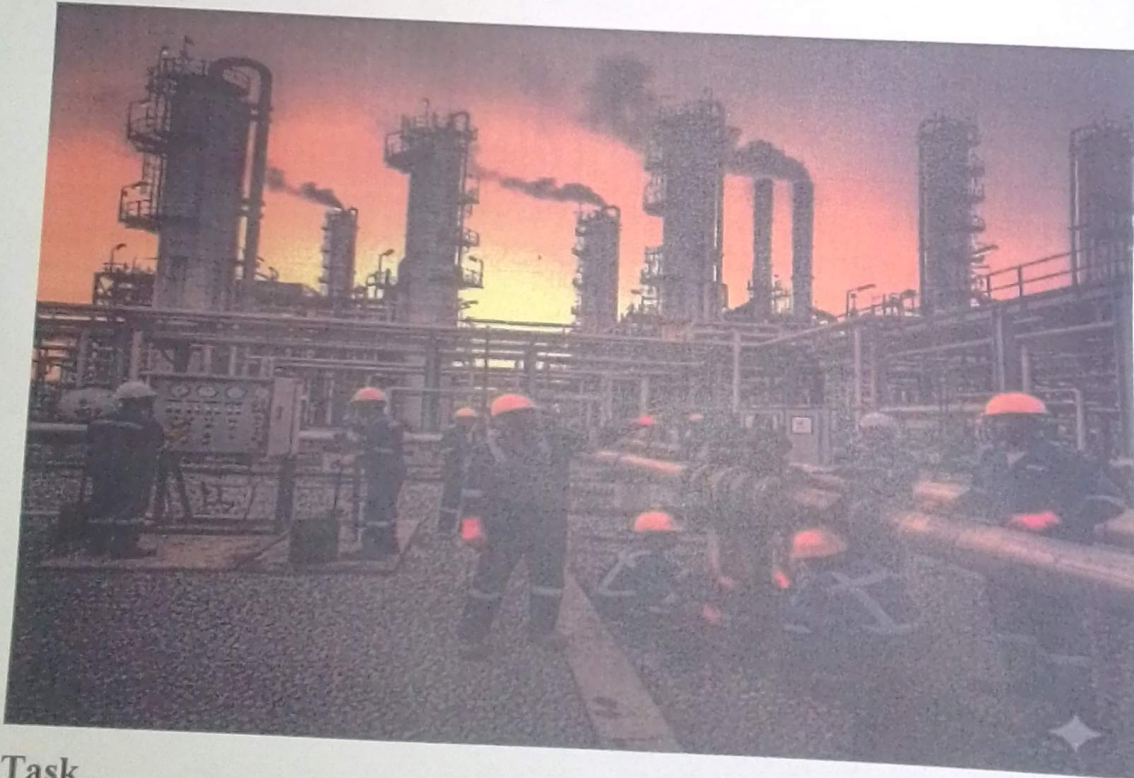
## Part II

*(Attempt one item from this part)*

### Item 5

Uganda is actively pursuing the development of its first crude oil refinery to ensure domestic energy security. The refinery is planned to be built in Kabaale village, Buseruka Sub-county in Hoima district near the rich crude oil fields.

However, residents are concerned about the potential environmental damage from refinery process particularly on air. Official from ministry of energy have organized workshops to address the locals. You have been invited to address the locals.



### Task.

Prepare a write up you will use during the presentation.



**Item 6.**

Many water sources in several districts are adversely affected by large scale crop cultivation and rapid establishment of medium -sized factories. While these endeavors have boosted the incomes of many families, they have become a source of significant concern for the district officials. In response, sensitization workshops have been organized to address the situation. You have been invited to address the public.

**Task.**

Prepare a write up message you will use.

*End.*